ITA0608 – Machine Learning Lab Practicals

EXP 06:

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Aim:

To implement the Naïve Bayes Algorithm in Python and evaluate its performance using the Confusion Matrix and Accuracy.

Algorithm:

- Load the dataset and split it into training and testing datasets.
- Create and train the Naïve Bayes classifier using the training data.
- Predict the class labels for the testing data using the trained model.
- Evaluate the model by generating the Confusion Matrix and calculating the Accuracy.
- Display the predicted results and performance metrics, and classify a new sample if required.

Program:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=42

)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

accuracy = accuracy_score(y_test, y_pred)

print("Confusion Matrix:\n")

print(cm)
```

```
print("\nAccuracy:", round(accuracy * 100, 2), "%")
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
new_sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_sample)
print("\nNew Sample:", new_sample[0])
print("Predicted Class:", iris.target_names[prediction][0])
```

Output:

```
... Confusion Matrix:

    [[19  0  0]
     [ 0 12  1]
     [ 0  0 13]]

    Accuracy: 97.78 %

    Classification Report:

                  precision    recall  f1-score   support

          setosa       1.00      1.00      1.00        19
      versicolor       1.00      0.92      0.96        13
       virginica       0.93      1.00      0.96        13

        accuracy                           0.98        45
       macro avg       0.98      0.97      0.97        45
    weighted avg       0.98      0.98      0.98        45


    New Sample: [5.1, 3.5, 1.4, 0.2]
    Predicted Class: setosa
```

Result:

The Naïve Bayes Algorithm was successfully implemented in Python using the Iris dataset.